

THE MAGMOID OF NORMALIZED STOCHASTIC KERNELS

(j.w.w. Elena Di Lavore and Márk Széles)

MARIO ROMÁN

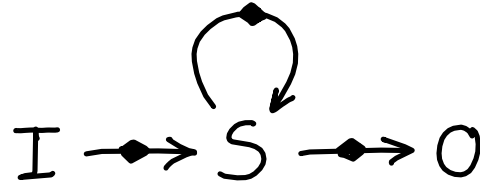
HIGHLIGHTS'26 — WIEN, 8th September.

NORMALIZED PROBABILITY DISTRIBUTIONS ARE NOT ASSOCIATIVE

- ① Formally, with category theory.
- ② Example, with a partially defined MDP.
- ③ Applied, to probabilistic functional programming syntax.

①. Formally, with category theory.

MONADS



Automata with effects.

```
(do x ← dice  
    y ← dice  
    return (x+y))
```

Functional prog. with effects

Powerset monad

NFA

Relational queries

Identity monad

DFA

Functional programs

Maybe monad

Partially defined DFA

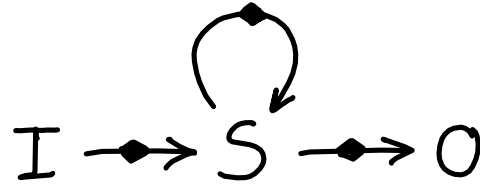
Error-handling programs

Distribution monad

MDP

Probabilistic programs

MONADS



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Normalized distribution monad?

Partially defined MDP

Probabilistic error-handling progs.

MIXING STOCHASTICITY OVER PARTIALITY

PROPOSITION (Distribution monad).

$$DX = \left\{ \sum_{i=0}^n \lambda_i |x_i\rangle \mid \sum_{i=0}^n \lambda_i = 1, x_i \in X, \lambda_i \in \mathbb{R}^+ \right\}.$$

PROPOSITION (Maybe monad).

$$MX = X + \{\perp\}.$$

CONJECTURE (Normalized distribution monad?).

$$MDX \cong \left\{ \sum_{i=0}^n \lambda_i |x_i\rangle \mid \sum_{i=0}^n \lambda_i = 1 \text{ or } 0, x_i \in X, \lambda_i \in \mathbb{R}^+ \right\}.$$

MIXING STOCHASTICITY OVER PARTIALITY

The resulting algebraic theory is not associative.

COUNTEREXAMPLE.

$$\begin{aligned} \frac{1}{2}|t_1\rangle + \frac{1}{2}|t_2\rangle & \left[t_1, t_2 \setminus \frac{1}{3}|x\rangle + \frac{2}{3}|z\rangle, \frac{2}{3}|y\rangle + \frac{1}{3}|z\rangle \right] [z \setminus \perp] \\ &= \frac{1}{6}|x\rangle + \frac{1}{3}|y\rangle + \frac{1}{2}|z\rangle [z \setminus \perp] = \frac{1}{3}|x\rangle + \frac{2}{3}|y\rangle \end{aligned}$$

$$\begin{aligned} \frac{1}{2}|t_1\rangle + \frac{1}{2}|t_2\rangle & \left[t_1, t_2 \setminus \frac{1}{3}|x\rangle + \frac{2}{3}|z\rangle [z \setminus \perp], \frac{2}{3}|y\rangle + \frac{1}{3}|z\rangle [z \setminus \perp] \right] \\ &= \frac{1}{2}|t_1\rangle + \frac{1}{2}|t_2\rangle [t_1, t_2 \setminus |x\rangle, |y\rangle] = \frac{1}{2}|x\rangle + \frac{1}{2}|y\rangle \end{aligned}$$

MIXING STOCHASTICITY OVER PARTIALITY

THEOREM [Sokolova, Woracek, 2018; Thm 5.3]. There is a single functorial one-point extension of convex algebras.

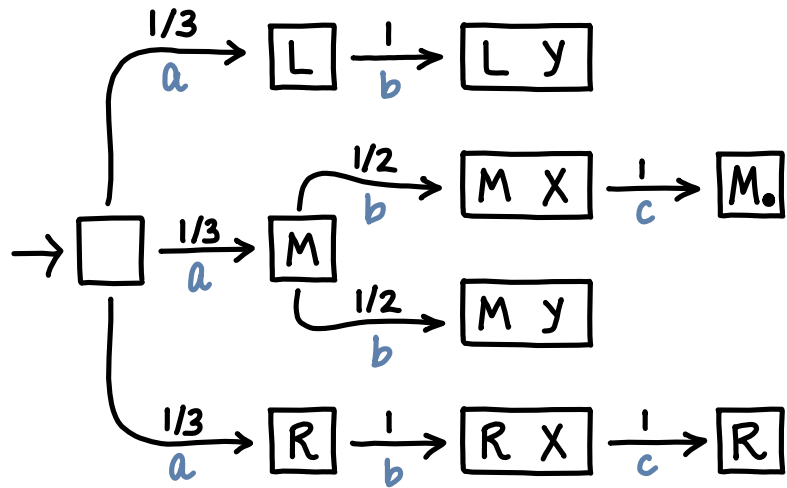
THEOREM [DRS26]. No mixture of the distribution monad ($\mathcal{D}$) over the maybe monad ($\mathcal{M}$) other than the trivial one.

HIGHLIGHT. We accept that normalized semantics is not associative: a non-associative monad.

1. Explicitly, with a partially-defined MDP.

TWO INTERPRETATIONS

(This is equivalent to the Monty Hall problem)



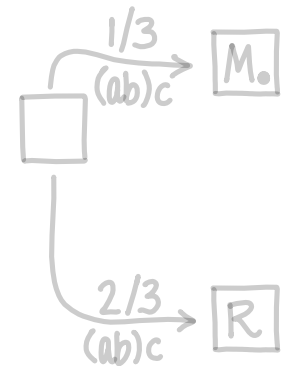
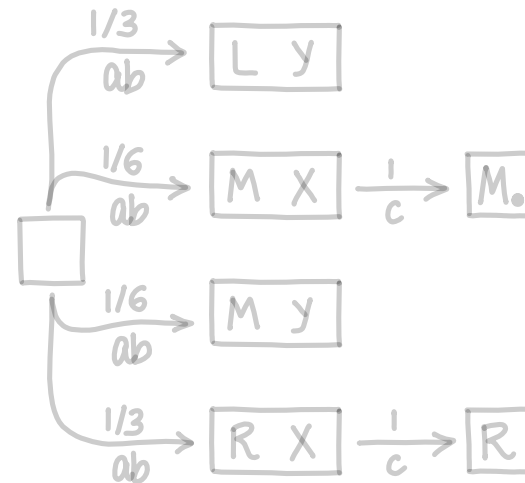
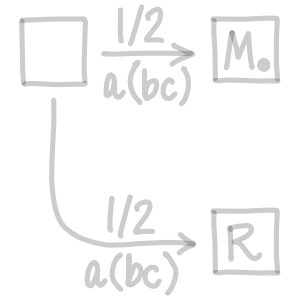
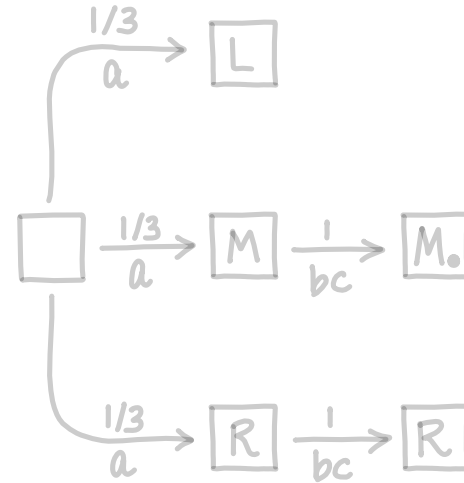
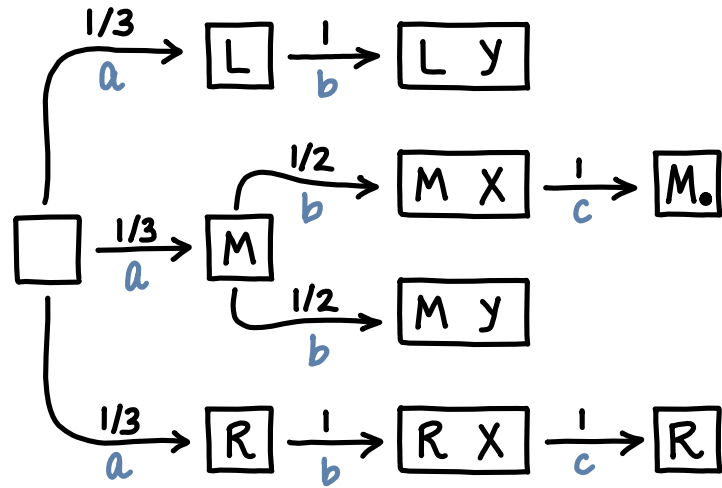
Consider a partially defined MDP. For each letter and each state we either get a distribution or nothing.

How to run abc ?

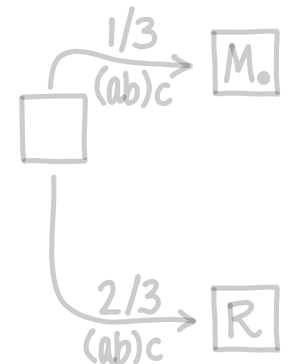
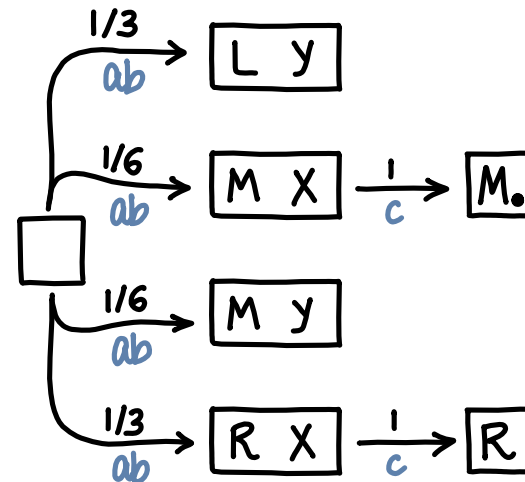
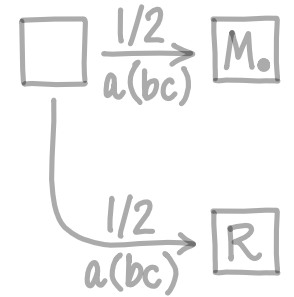
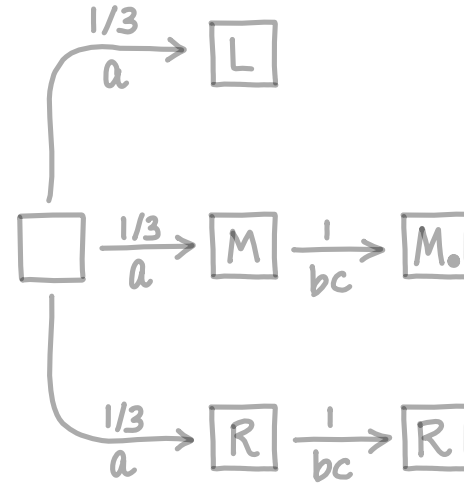
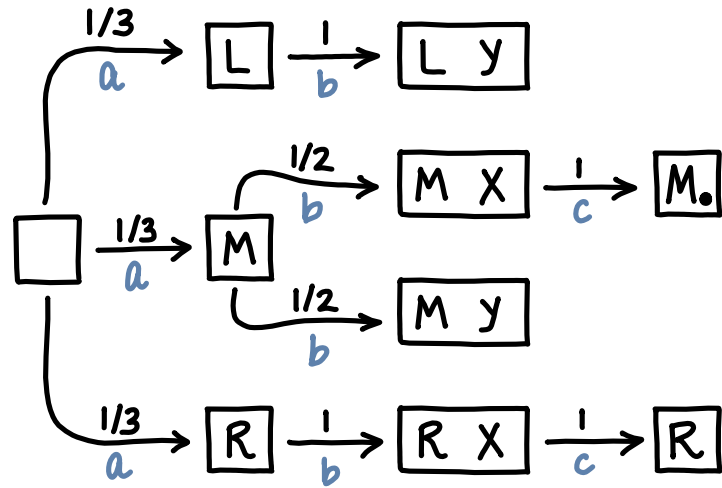
It depends whether we mean

$(ab)c$ versus $a(bc)$.

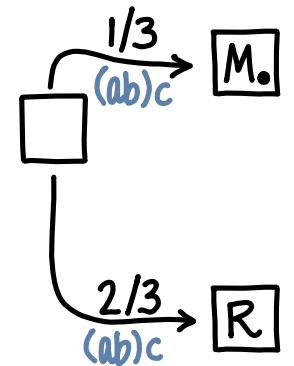
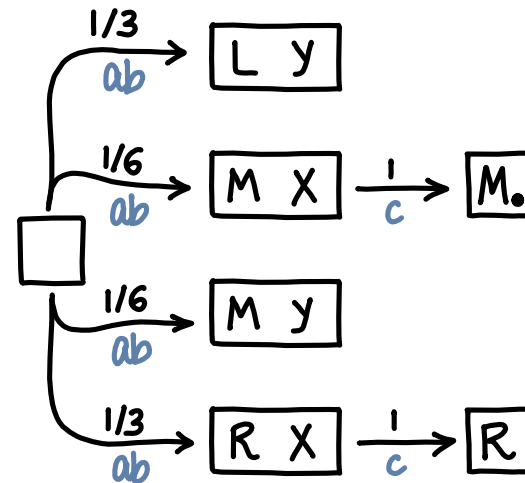
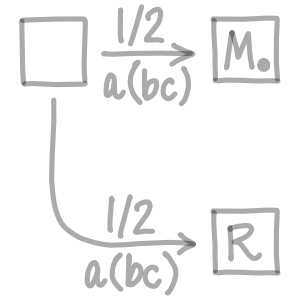
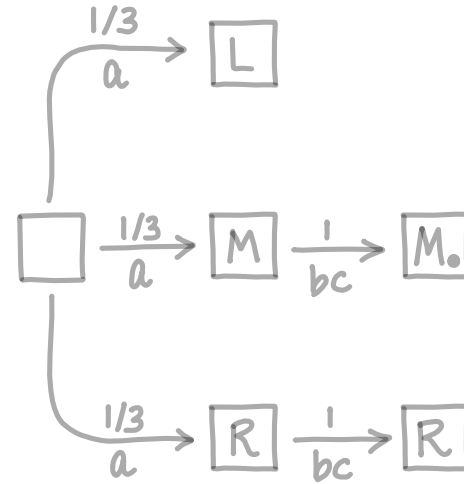
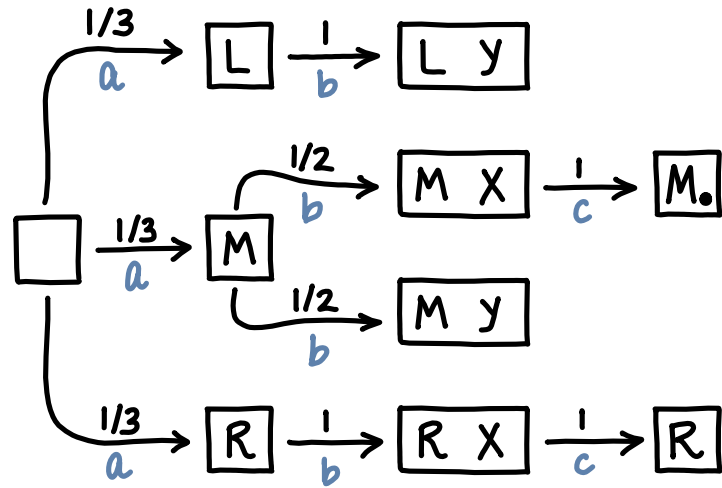
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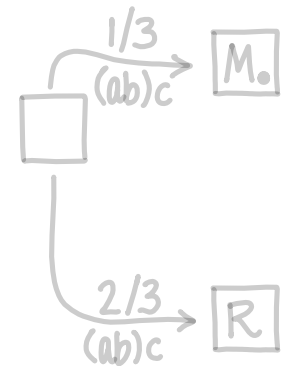
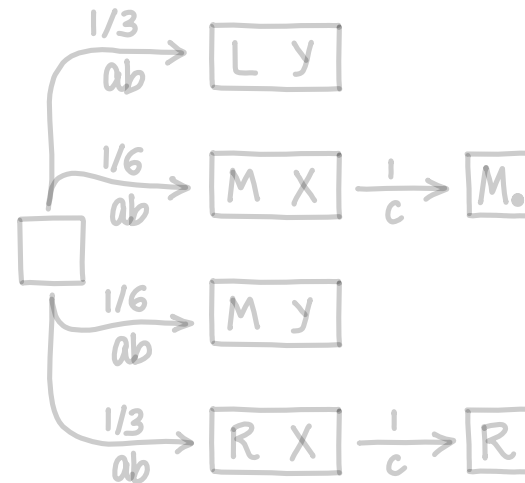
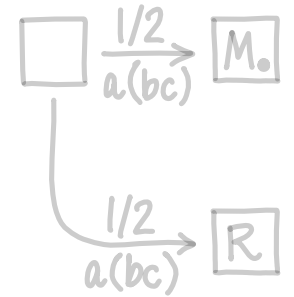
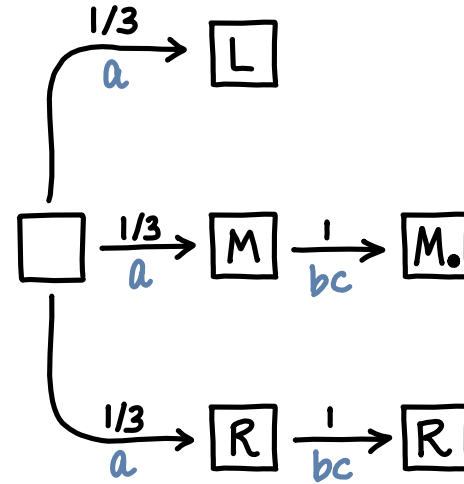
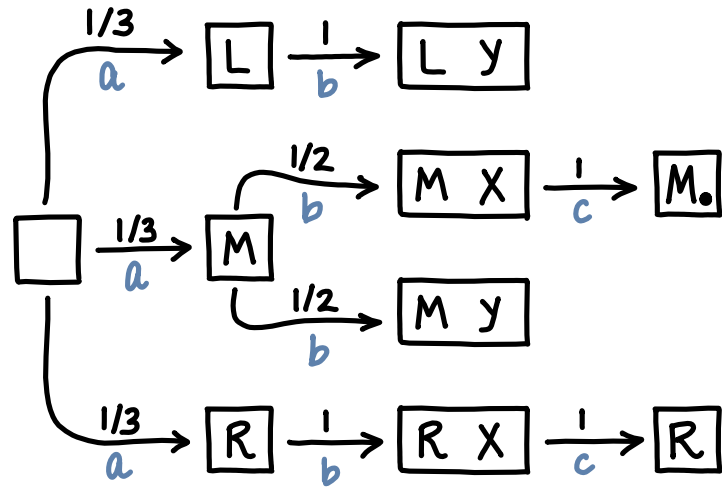
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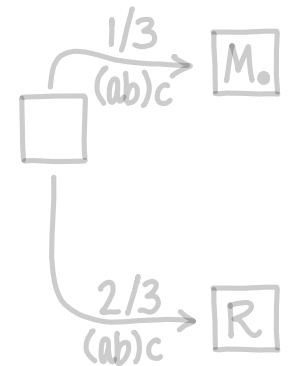
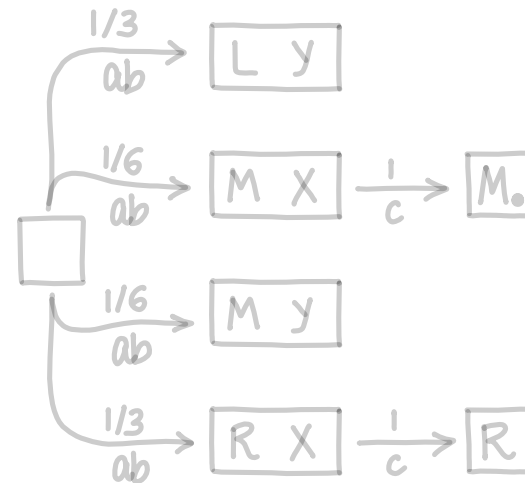
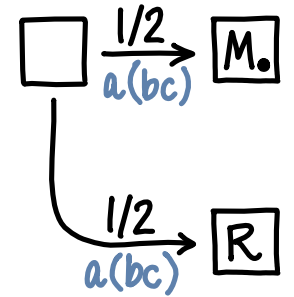
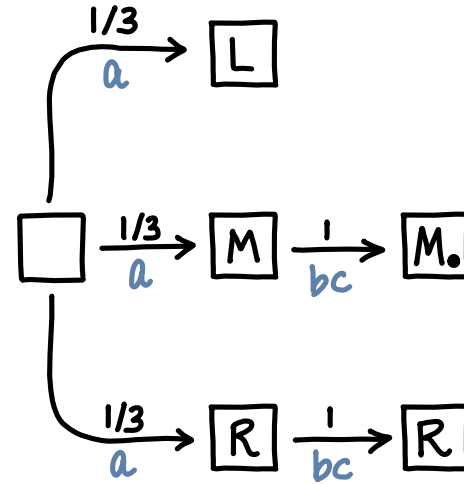
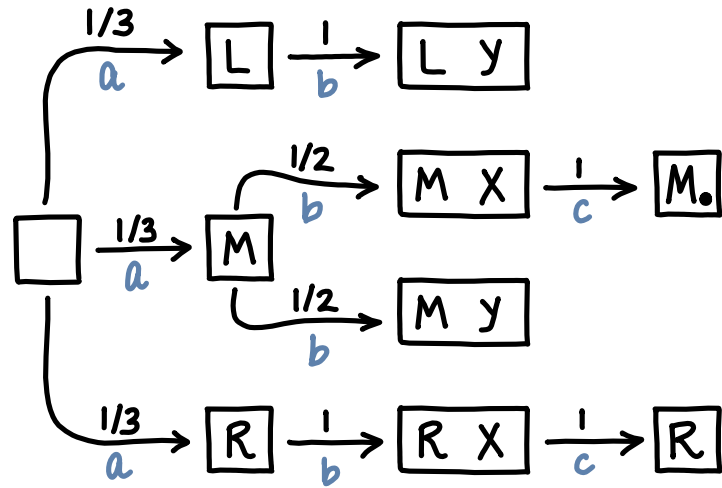
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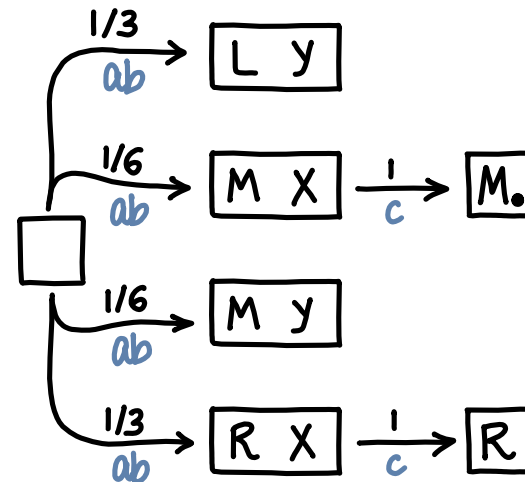
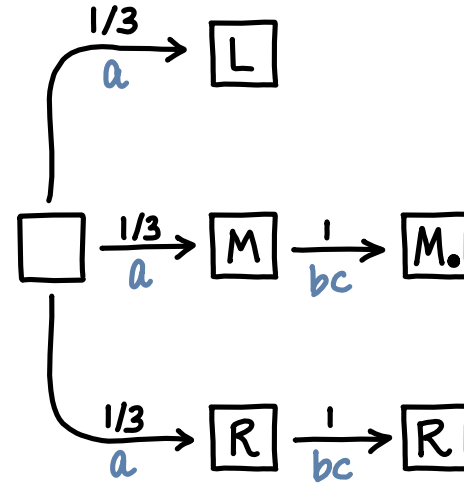
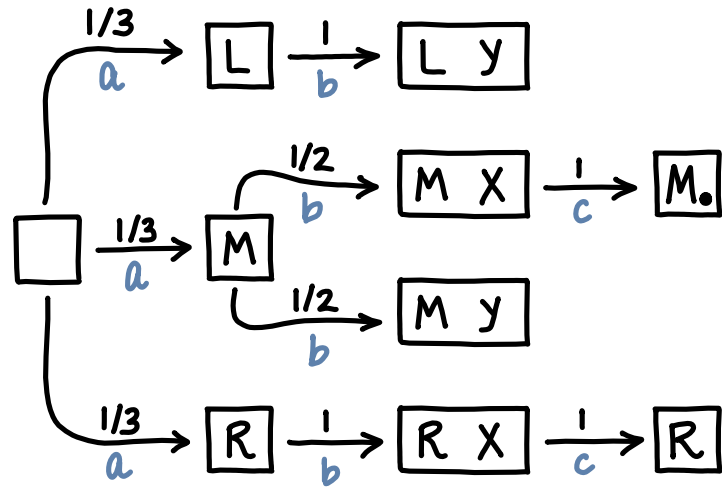
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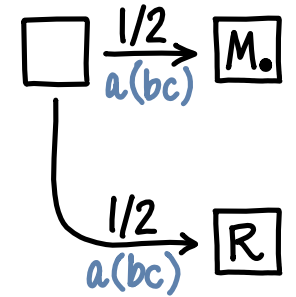
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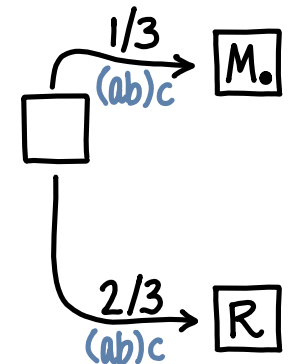
TWO INTERPRETATIONS



Naive filter



Bayes' update



MULTIPLE INTERPRETATIONS

In general, given a partially defined MDP with transitions

$$\gamma_a : S \longrightarrow \text{MDS}, \quad \text{for } a \in \Sigma;$$

we get not only a transition function for each word,

$$\gamma_w : S \longrightarrow \text{MDS}, \quad \text{for } w \in \Sigma^*;$$

but a transition function for each parenthesized word,

$$\gamma_t : S \longrightarrow \text{MDS}, \quad \text{for } t \in \text{Bintree}(\Sigma).$$

2. Beyond : programming

NON-ASSOCIATIVE METALANGUAGE

Associative probabilistic programming
allows us to query
CONDITIONAL DISTRIBUTIONS.

```
(do (illness treatment outcome) ← data  
    () ← (observe treatment 'X)  
    return (outcome))
```

“what are the odds of remission for
a patient assigned to treatment X?”

Non-Associative probabilistic programming
moreover allows us to query
INTERVENTIONAL DISTRIBUTIONS.

```
(do (illness _ ) ← data  
    (outcome) ← (do (illness treatment outcome) ← data  
        () ← (observe treatment 'X)  
        () ← (observe illness illness)  
        return (outcome))  
    return (outcome))
```

“what are the odds of remission if we
give treatment X to the patient?”

END

- Di Lavore, Román. Evidential Decision Theory via Partial Markov Categories (2023).
- Di Lavore, Román, Széles. Normalized Probabilistic Semantics is Not Associative (2025).
- Román. Causal Probabilistic Programming via Magmatic Do-Notation (2026).